Example: Solar Neutrino

Consider solar neutrino with energy 1 MeV. How far it can travel without interactions?

1. Cross-section on electrons:

$\boxed{\mathrm{MeV} = 10^6\,\mathrm{eV}}$

$$\sigma(\nu_e e \to \nu_e e) \approx 10^{-44}\,\mathrm{cm}^2$$

2. Number density of targets:

$$n = \rho \cdot N_A \cdot \frac{Z}{A} = 1.4\frac{\mathrm{g}}{\mathrm{cm}^3} \cdot \frac{6.02 \cdot 10^{23}}{\mathrm{mole}} \cdot \frac{0.86}{\mathrm{g/mole}} \approx \frac{7 \cdot 10^{23}}{\mathrm{cm}^3}$$

3. Interaction length:

$$\lambda = \frac{1}{n \cdot \sigma} = \frac{\mathrm{cm}^3}{7 \cdot 10^{23}} \cdot \frac{1}{10^{-44}\mathrm{cm}^2} \approx 10^{20}\,\mathrm{cm} \approx 10^9\,R_\odot$$

Example: PeV Neutrino in the Earth